

Luv Patel

+91-6355961895 | luvvvpatel@gmail.com | luvvv.me | [linkedin/luvv](https://www.linkedin.com/in/luvv) | [github/luvp21](https://github.com/luvp21)

Education

Nirma University, Ahmedabad | B.Tech in Computer Science and Engineering

July 2023 – July 2027 (Expected)

- CGPA: 8.42 / 10.0

Work Experience

Befree Global | AI Intern, COE

May 2026 – July 2026

Built an end-to-end bank statement processing platform on the CoE AI Innovation team using FastAPI, Next.js, and PostgreSQL.

- Designed a three-tier cascade classifier (Rule Engine → TF-IDF/LR → Gemini LLM) with a self-improving RuleLearner that shifts 40% of LLM calls to O(1) keyword lookups, cutting API costs by 85% and p95 latency to under 1.5s.
- Built a coordinate-locked PDF extraction engine using PyMuPDF to detect header geometry and project column boundaries across pages, achieving sub-cent balance reconciliation accuracy on multi-format bank statements.
- Architected a modular FastAPI backend with JWT auth, session revocation, RBAC, and immutable audit logging; built a Next.js 15 HITL review dashboard with a four-phase enrichment pipeline processing 10,000+ monthly transactions.

Projects

Crux | Real-Time Collaborative DSA Platform | [GitHub](#) [Live Demo](#)

- **Technologies:** TypeScript, Next.js 14, PartyKit WebSockets, Neon Postgres, Drizzle ORM, Auth.js v5, Judge0, Gemini API
- Architected a room-based DSA platform using Next.js 14, Neon Postgres, and Auth.js v5, decoupling SSR page loads from real-time PartyKit WebSocket streams
- Built an automated code execution pipeline with Judge0, engineering dynamic AST/string driver generators for Python, JS, C++ , and Java to serialize stdin/stdout test cases and aggregate multi-case verdicts.
- Designed a centralized solution privacy gate (canViewSolution) enforcing crew membership and solve verification; built a dual-layer streak engine with optimistic updates and midnight cron reconciliation.

BHOOMI | Satellite Environmental Forecasting Platform | [GitHub](#) [Live Demo](#)

- **Technologies:** Python, PyTorch, React 18, Express, PostgreSQL, Leaflet Canvas, XGBoost, Prophet, SARIMA, Gemini Flash
- Engineered a multi-model ML pipeline across 1,183 spatial grid cells (3km × 3km), using PyTorch BiLSTM with Pinball Quantile Loss for NDVI/soil moisture, SARIMA for NO₂, and XGBoost for drought risk.
- Built a spherical wind-vector causal engine (Haversine & great-circle azimuths) computing upwind green buffer boundary cells; rendered heatmaps via a Leaflet HTML5 Canvas overlay at 60 FPS.
- Integrated Google Gemini Flash to automatically synthesize empirical satellite telemetry and causal attribution models into print-ready municipal executive PDF briefings.

AI Voice Ordering Copilot | [GitHub](#)

- **Technologies:** Python, FastAPI, Fastify, WebSockets, Silero VAD, Deepgram STT, Gemini Flash, PostgreSQL, Petpooja POS
- Engineered a real-time voice ordering agent over WebSockets using a streaming pipeline (Silero VAD → Deepgram STT → Gemini Flash → TTS), cutting end-to-end voice latency down to sub 3s.
- Implemented a RapidFuzz fuzzy matching layer to normalize spoken orders into canonical menu items, preventing LLM dish hallucinations and committing verified Kitchen Order Tickets (#KOT) to Petpooja POS.
- Designed an in-process session cache with background async PostgreSQL flushes to serve active call turns, bypassing Neon serverless DB cold-start latencies (300–500ms).

Achievements

Amazon ML Summer School 2026 | *(Certificate)* | 8-module program covering Supervised/Unsupervised Learning, DNNs, RL, GenAI & LLMs, and Causal Inference.

Breach Hackathon | Winner *(Certificate)* | EV rental platform with QR access, wallet payments, and station locator.

Aetrix Hackathon | Runner Up *(Certificate)* | Satellite platform for environmental monitoring and climate risk analysis.

HACKaMINeD 2026 | Runner Up *(Certificate)* | AI-powered revenue and voice copilot platform for restaurants.

GUVI-HCL Hackathon | National Finalist *(Certificate)* | Agentic Honeypot; attended India AI Summit 2026.

Hackspllosion 2026 | National Top 60, Level 2 Cleared *(Certificate)* | Built AI resume-job matching platform on GenW.AI.

Leadership & Activities

Computer Society of India | General Secretary *(Website)*

January 2025 – Present

- Built and scaled an event platform handling 1.1M+ requests and 50K+ users with 98% uptime; designed a modular backend and database schema managing 1,500+ registrations.
- Coordinated logistics and operations for Cubix'25, HackNUthon 6.0, and Hackamined 2026, serving 3,400+ participants.

Skills

Languages: C++ , Python, Java, TypeScript, JavaScript, SQL

Frameworks & Libraries: React, Next.js, Node.js, Express, Fastify, FastAPI, Flask, Prisma ORM, Pydantic

Distributed Systems & Databases: PostgreSQL, MongoDB, WebSockets, REST APIs, JWT Auth, pgvector

AI/ML: PyTorch, Scikit-learn, HuggingFace, LangChain, Google Gemini API, FAISS, RAG, LLM Agents

Tools & Practices: Docker, Git, Linux, AWS (Familiar), Vercel, Agile, System Design, Postman, Turborepo

Coursework: Data Structures & Algorithms, OOP, Design & Analysis of Algorithms, Operating Systems, AWS, DBMS, ML